International Journal of Heat and Mass Transfer 149 (2020) 119251

Contents lists available at ScienceDirect

International Journal of Heat and Mass Transfer

journal homepage: www.elsevier.com/locate/hmt

Effect of a flow behavior on the thermal performance of closed-loop and closed-end pulsating heat pipes

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ARTICLE INFO

Article history:
Received 5 September 2019
Revised 29 November 2019
Accepted 12 December 2019
Available online 24 December 2019

ABSTRACT

In this study, the effect of a flow behavior is taken into account to determine which type of a pulsating heat pipe (PHP) performs better between a closed-loop pulsating heat pipe (CLPHP) and a closed-end pulsating heat pipe (CEPHP). The thermal performances of a CLPHP and a CEPHP are experimentally evaluated and compared. For this, MEMS techniques are used to fabricate silicon-based PHPs with ten turns. Pyrex glass covers the etched silicon wafer to visualize the flow behavior inside the PHPs. The PHPs have rectangular channels with a hydraulic diameter of 923μm. Ethanol and R-134a are used as the working fluids. Based on the experimental data, a criterion of a diameter ratio ($D_h/D_{h,\,\mathrm{crit}}$) is proposed to quantify the effect of a flow behavior on the thermal performance of two types of PHPs. A CEPHP performs better than a CLPHP when $D_h/D_{h,\,\mathrm{crit}} < 0.5$. In the case of PHPs filled with ethanol ($D_h/D_{h,\,\mathrm{crit}} = 0.29$), the CEPHP shows 33% lower thermal resistance and 20% higher maximum allowable heat flux than the CLPHP. In this case, an oscillation mode is observed inside both types of PHPs. In contrast, a CLPHP performs better than a CEPHP when $D_h/D_{h,\,\mathrm{crit}} > 0.5$. In the case of PHPs filled with R-134a ($D_h/D_{h,\,\mathrm{crit}} = 0.69$), the CLPHP shows 72% lower thermal resistance and 117% higher maximum allowable heat flux than the CEPHP. In this case, a circulation mode is observed inside the CLPHP, while an oscillation mode is observed inside the CEPHP. To explain why one type of PHP performs better than the other, an average volumetric fraction is introduced. An average volumetric fraction in the condenser section is experimentally shown to be a major contributor to the thermal performance; an increase of the average volumetric fraction in the condenser section leads to an improvement of the thermal performance due to enhanced latent heat transfer in the condenser section.

1. Introduction

With the increasing demand for densely packed electronic systems, thermal problems in electronics have increased significantly. Therefore, an effective thermal management solution is required for cooling electronic devices and systems. A pulsating heat pipe (PHP), invented by Akachi et al. [1], is a promising cooling solution because of its high thermal performance and geometrical simplicity. Generally, a PHP can be classified into two types depending on its channel configuration [2–5]: a closed-loop PHP (CLPHP) and a closed-end PHP (CEPHP). In the case of a CLPHP, a serpentine channel forms a single closed loop. On the other hand, in the case of a CEPHP, both ends of a serpentine channel are disconnected to each other.

To figure out which type of a PHP performs better between a CLPHP and a CEPHP, many researchers have compared the thermal performance of a CLPHP and a CEPHP [6–11]. Yang et al. [10] claimed that the CLPHP has a higher thermal performance than the CEPHP due possibly to a circulation mode inside a CLPHP, which is simply based on their conjecture. On the other hand, Jun and Kim [11] reported that the CEPHP has a higher thermal performance than the CLPHP and that a circulation mode was not visually observed inside their CLPHPs. To resolve the contradictory results, it is postulated that a flow behavior inside a CLPHP is a key factor to determining which type of a PHP performs better between two types of PHPs based on the fact that the thermal performance of a CLPHP is strongly coupled with a flow behavior inside a CLPHP [12,13]. Therefore, it is necessary to consider the effect of a flow behavior inside a CLPHP when two types of PHPs are compared.

The purpose of this study is to figure out under what condition a CLPHP performs better than a CEPHP and vice versa. For this, the thermal performances of CLPHPs and CEPHPs are experimentally compared considering the effect of a flow behavior inside a CLPHP. Using MEMS techniques, ten-turn PHPs, which have a serpentine

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https://doi.org/10.1016/j.ijheatmasstransfer.2019.119251

Nomenclature

A	area [m^2]
D	diameter [m]
g	gravitational acceleration [m/s^2]
Q	input power [W]
R_{th}	thermal resistance [K/W]
$\overline{T}$	average temperature [K]
V	volume [m^3]

Greek symbols

β	volumetric fraction [-]
ρ	density [kg/m^3]
σ	surface tension [N/m]

Subscripts, Superscripts

l	liquid
v	vapor
e	evaporator
a	adiabatic
c	condenser
vp	vapor plug
h	hydraulic
crit	critical
in	inlet

channel, are fabricated by etching a silicon substrate. For flow visualization, Pyrex glass is bonded to the etched silicon wafer. The hydraulic diameter of the channel is 923 μm. Ethanol and R-134a are used as a working fluids. High-speed photography and thermometry are used to investigate the flow and thermal characteristics of the CLPHP and the CEPHP. Based on the experimental data, a new design guideline is suggested to help thermal engineers to choose a better PHP between a CLPHP and a CEPHP.

2. Experiments

2.1. Fabrication of PHPs

To compare the flow and the thermal characteristics of a CLPHP and a CEPHP, MEMS techniques are used to fabricate ten-turn CLPHPs and CEPHPs. The PHP has a serpentine channel, which are engraved on a silicon wafer with a thickness of 1 mm using deep the reacting ion etching (DRIE) process. In the case of the CLPHP, a serpentine channel forms a single closed loop, as shown in Fig. 1(a). On the contrary, in the case of the CEPHP, both ends of a serpentine channel are not connected to each other, as shown in Fig. 1(b). In this study, the serpentine channel consists of an array of 20 vertical channels. The width and height of the channels are 2 mm and 0.6 mm, respectively. To visualize the flow behavior inside the PHP, Pyrex glass with a thickness of 0.7 mm covers an etched channel. The PHP has a length of 80 mm, a width of 53.5 mm, and a thickness of 1.7 mm. The lengths of the condenser, the adiabatic and the evaporator section are set to 24 mm, 32 mm, and 24 mm, respectively, as shown in Fig. 2. The degassing and charging of the working fluid into the PHP are conducted through two holes. In the present study, the fabrication process is identical to the process suggested by Youn and Kim [14].

Generally, the hydraulic diameter of a channel should be smaller than the critical diameter of a PHP. The critical diameter of a PHP, $D_{h,\,\text{crit}}$, is defined as follows [2–4]:

$$D_{h,\text{crit}} = 2\sqrt{\frac{\sigma}{g(\rho_l - \rho_v)}} \tag{1}$$

where σ,g,ρ_l, and ρ_v are surface tension, gravitational acceleration, density of liquid, and density of vapor, respectively. Under the constraint of the critical diameter of a PHP, Liu et al. [13] experimentally observed that as the hydraulic diameter of the channel increases, the flow behavior of the CLPHP changes from an oscillation mode to a circulation mode in a vertical orientation under a bottom-heating condition. Moreover, a loop thermosyphon, which has a larger hydraulic diameter of the channel than the critical diameter of a PHP, is known to operate in a circulation mode in a vertical orientation under a bottom-heating condition [14–16]. Therefore, it can be postulated that a flow behavior inside a CLPHP in a vertical orientation under a bottom-heating condition will change from an oscillation mode to a circulation mode as value of $D_h/D_{h,\,\text{crit}}$ increases. Fig. 3 depicts $D_h/D_{h,\,\text{crit}}$ of ethanol and R-134a in the present study. R-134a has a much larger value of $D_h/D_{h,\,\text{crit}}$ than ethanol. Therefore, ethanol and R-134a are used as working fluids to compare the flow behavior inside a CLPHP depending on

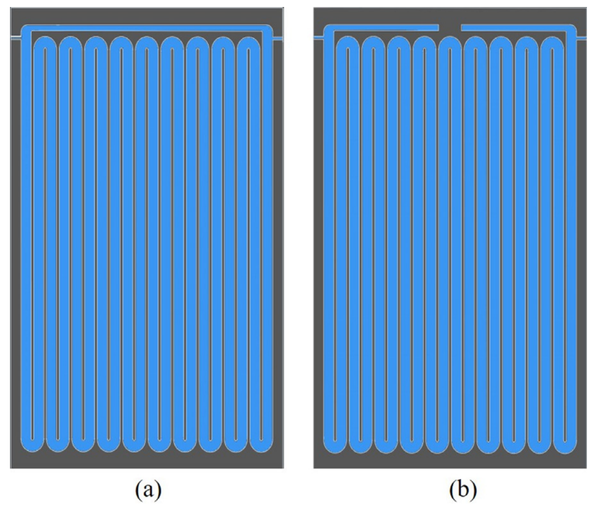

Fig. 1. Schematic of the (a) CLPHP and the (b) CEPHP.

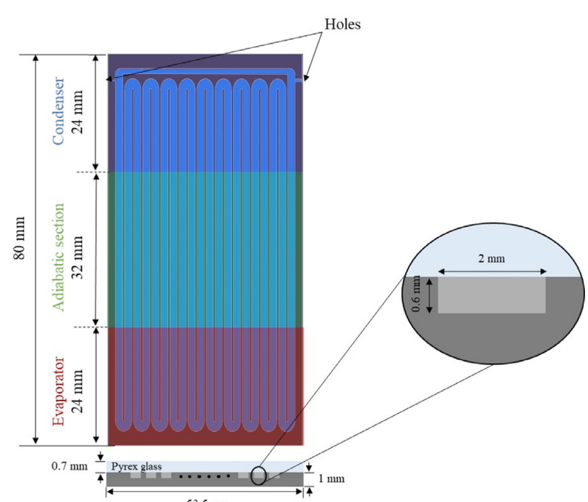

Fig. 2. Schematic and dimensions of the fabricated PHPs.

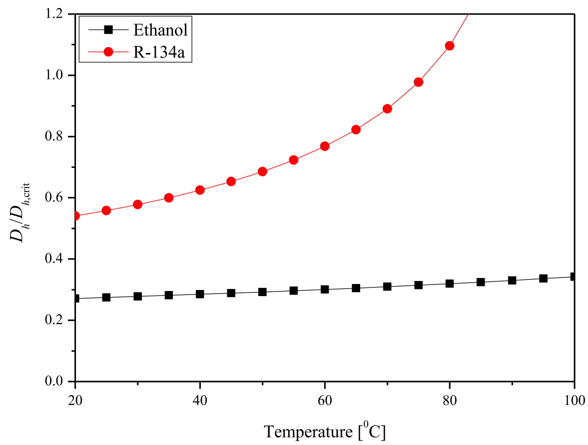

Fig. 3. The critical diameter of the various working fluids.

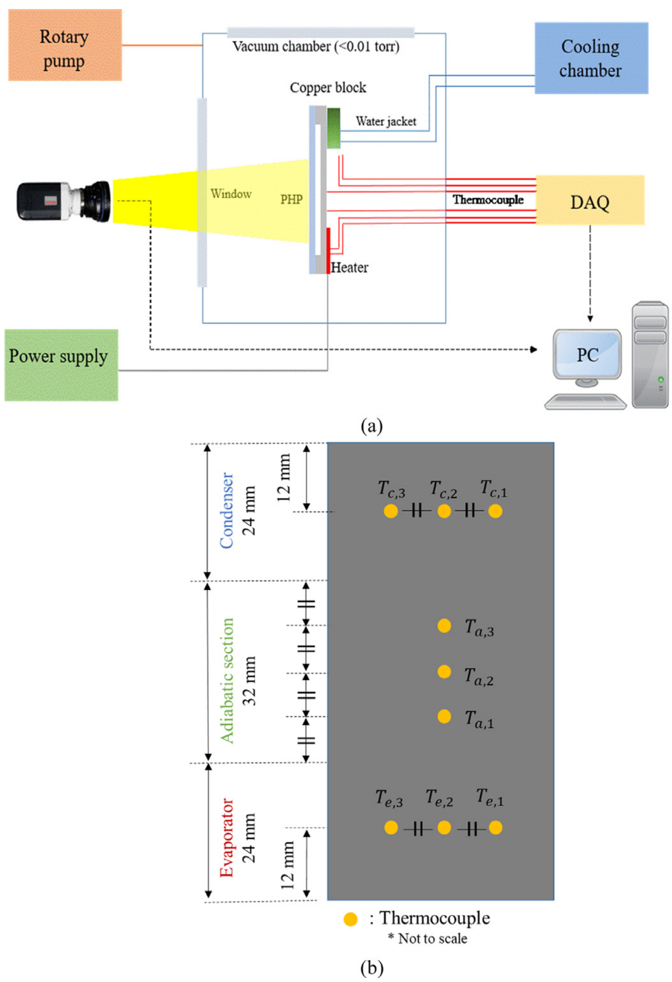

Fig. 4. Overall schematic of (a) experimental setup and (b) temperature measurement points.

the $D_h/D_{h,\,\mathrm{crit}}$. Every experiment is performed in a vertical orientation under a bottom-heating condition. Except for the channel configuration and the working fluid of a PHP, every experimental parameter is maintained to have the same value. Table 1 summarizes all experimental parameters used in this study.

2.2. Experimental setup

A schematic of the experimental setup is presented in Fig. 4(a). To minimize the heat loss by convection, every experiment is conducted in a vacuum chamber. A thin film of titanium is evenly deposited on the silicon surface of the evaporator section using a DC magnetron sputtering system (SRN-110, Sorona Inc.). The thin film of titanium is heated using a DC power supply (N5772A, Agilent Technologies). A bath circulator supplies the coolant, ethanol, into a copper block for cooling the condenser section. The inlet temperature of the condenser section is constantly maintained at 10°C. As shown in Fig. 4(b), K-type thermocouples are attached to the silicon surface of the PHP for measuring the surface temperatures of the PHP. The accuracy of the thermocouples is estimated to be within ± 0.4%. The measured temperature data are saved on a personal computer through the data acquisition system (34980A, Agilent Technologies). In this study, the maximum temperature of PHP is limited to 100°C, which is typically imposed as the maximum allowable temperature for electronic devices. In the case of PHPs filled with R-134a, the maximum temperature is limited to 55°C due to a high saturation pressure of R-134a, as shown in Appendix A. Every experiment is s when the evaporator temperature exceeds the maximum allowable temperature. Hence, the thin film heater is heated by the power supply from 4 W by an increment of 4 W until the evaporator temperature reaches the maximum allowable temperature. Moreover, a high-speed camera (M320S, Phantom) is used to figure out the flow behavior inside the PHP.

To evaluate the thermal performance of the PHP, the thermal resistance of the PHP, R_{th}, is defined as follows:

$$R_{th} = \frac{\overline{T_e} - \overline{T_c}}{Q_{in}} \tag{2}$$

where $\overline{T_e}$, $\overline{T_c}$, and Q_{in} are the average temperatures of the evaporator and the condenser sections, and input power, respectively. The average temperatures included in the definition of the thermal resistance are evaluated using an arithmetic mean of temperatures, which are taken 20 min after the PHP enters the pseudo-steady state. The experimental uncertainty for calculating the thermal resistance is identical to Kim and Kim [15]. The overall uncertainty of the thermal resistance is estimated to be less than 6%.

3. Results and discussion

3.1. Thermal resistance of the CLPHP and the CEPHP

Fig. 5 shows the thermal resistance of PHPs filled with R-134a at various input powers. In the case of PHPs filled with R-134a, non-operation of the PHP is not observed. Moreover, the thermal resistance of the CLPHP is much lower than that of the CEPHP at various input powers. The CLPHP has up to 72% lower thermal

Table 1
All experimental parameters used in this study.

Hydraulic diameter [μm]	Configuration	Working fluid	Filing ratio	Input power [W]	Inclination angle (°)
923	Closed-loop	R-134a	50%	4~(Increment :4)	90 (Vertical)
	Closed-end				
	Closed-loop	Ethanol			
	Closed-end				

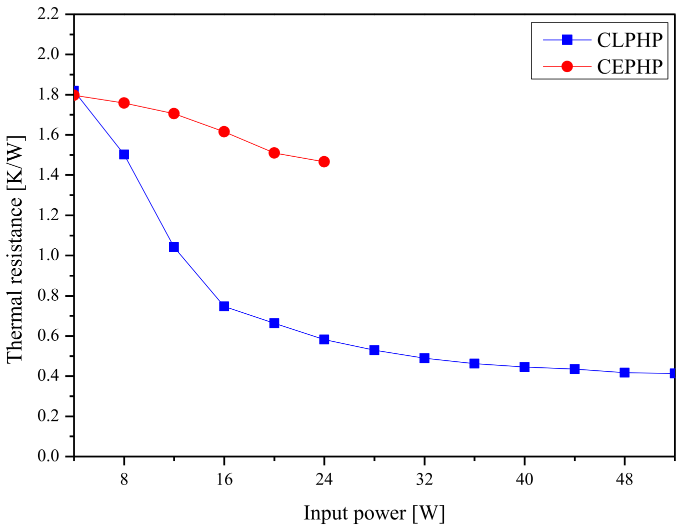

Fig. 5. The thermal resistance of the CLPHP and the CEPHP filled with R-134a at various input powers.

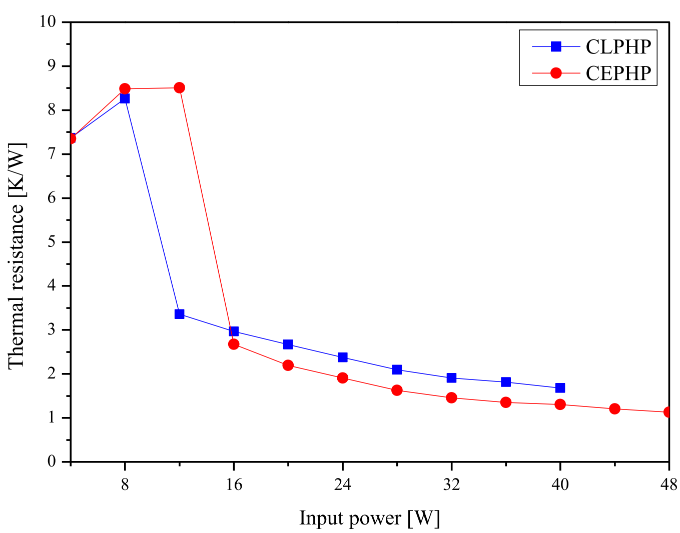

Fig. 6. The thermal resistance of the CLPHP and the CEPHP filled with ethanol at various input powers.

resistance than the CEPHP. Furthermore, the CLPHP shows 117% higher maximum allowable heat flux compared to the CEPHP.

In the case of PHPs filled with ethanol, different tendencies are observed compared to the case of PHPs filled with R-134a. Fig. 6 depicts the thermal resistance of PHPs filled with ethanol at various input powers. Unlike the above results of PHPs filled with R-134a, non-operation of the PHP is observed before startup. PHPs filled with ethanol require a larger input power for startup than PHPs filled with R-134a, because ethanol has a relatively higher heat of vaporization than R-134a. During non-operation of a PHP, heat is transferred through conduction with no fluid motion. Therefore, the thermal resistance of a PHP is relatively higher during non-operation of a PHP. After startup, in the case of PHPs filled with ethanol, the thermal resistance of the CEPHP is lower than that of the CLPHP. Moreover, the CEPHP has up to 33% lower thermal resistance than the CLPHP. Furthermore, the CEPHP shows 20% higher maximum allowable heat flux compared to the CLPHP. The percentage of reduction in the thermal resistance and the percentage of enhancement of the maximum allowable heat flux are similar to the experimental results reported by Jun and Kim [11].

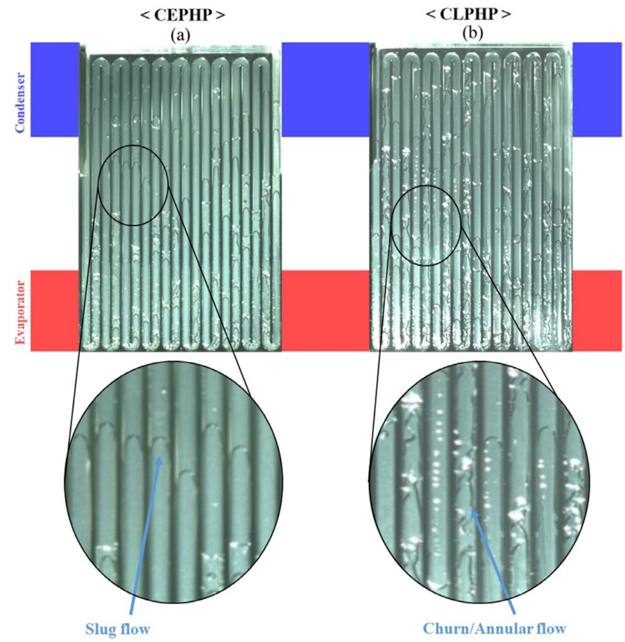

Fig. 7. Flow characteristics of (a) the CEPHP and (b) the CLPHP filled with R-134a at 24 W.

Additionally, in the case of the CLPHP filled with ethanol, earlier startup is observed than that of the CEPHP filled with ethanol [11].

3.2. Flow characteristics of the CLPHP and the CEPHP

The flow characteristics of the CLPHP and the CEPHP are investigated after startup of each PHP. Section 3.2.1 covers the flow characteristics of PHPs filled with R-134a, while Section 3.2.2 only covers the flow characteristics of PHPs filled with ethanol.

3.2.1. CLPHP operating in a circulation mode

Fig. 7 shows the flow characteristics of both types of PHPs at the input power of 24 W. In the case of the CEPHP, the flow pattern of vapor plugs is slug flow, as shown in Fig. 7(a). In contrast, in the case of the CLPHP, churn/annular flow is usually observed, as shown in Fig. 7(b). Furthermore, Fig. 8 depicts the temporal variation of liquid slugs and vapor plugs in the 10th and 11th channels of the CEPHP. The blue and red dashed lines in Fig. 8 represent the positions of the menisci in the 10th and 11th channels of the CEPHP, respectively. A circulation mode cannot be observed inside the CEPHP due to its channel configuration. Hence, the CEPHP is observed to operate in an oscillation mode. On the other hand, the CLPHP is observed to operate in a circulation mode, as presented in Fig. 9. The red, blue, and purple lines show the movements of three different menisci in the 11th channel of the CLPHP. Inside the 10th channel of the CLPHP, the vapor plugs move upward continuously, while the vapor plugs inside the 11th channel of the CLPHP move downward continuously. Moreover, in the case of the CLPHP, the vapor plugs move from one channel to another successively, while the vapor plugs oscillate in each vertical channel in the case of the CEPHP. Hence, in the case of the CLPHP, uni-directional circulating flow is observed. Moreover, big differences are observed in the condenser section of the CEPHP and the CLPHP. In the case of the CLPHP, vapor plugs are penetrated deeply into the condenser section compared to the case of CEPHP. To quantify the penetration rate of vapor plugs into the condenser section, vapor plugs and liquid slugs in the condenser section are traced. Moreover, the

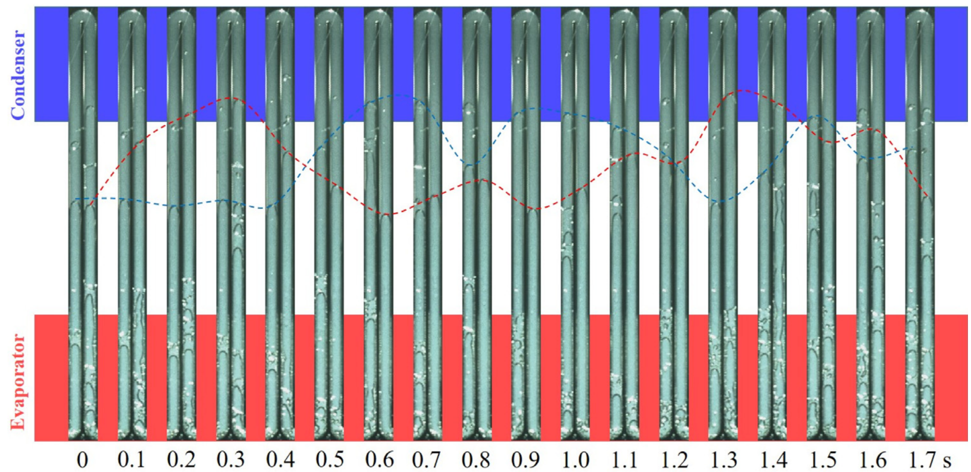

Fig. 8. Temporal variations of liquid slugs and vapor plugs in the 10th and 11th channels of the CEPHP filled with R-134a at 24 W.

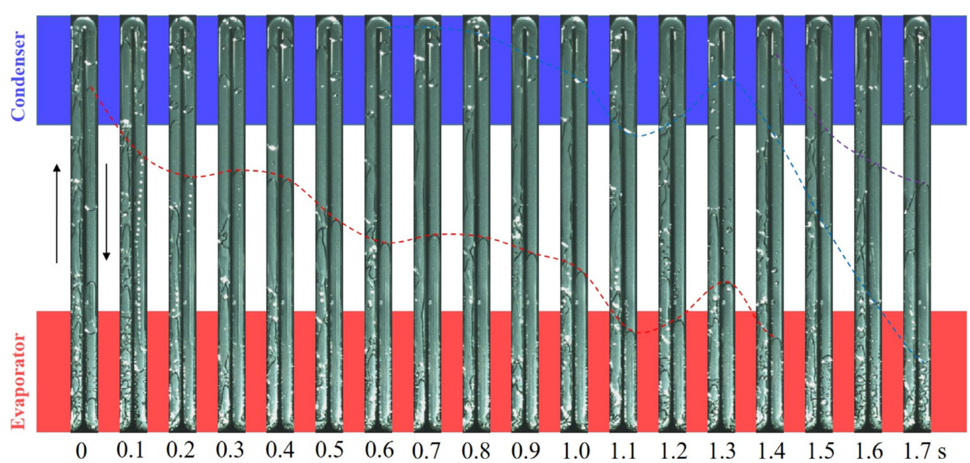

Fig. 9. Temporal variations of liquid slugs and vapor plugs in the 10th and 11th channels of the CLPHP filled with R-134a at 24 W.

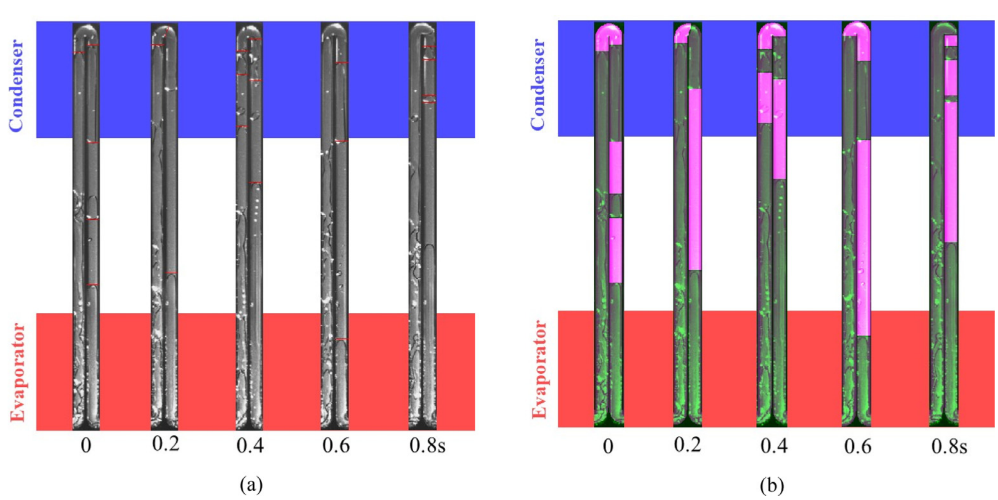

Fig. 10. Temporal identification of areas occupied by vapor plugs and liquid slugs in the 10th and 11th channels of the CLPHP filled with R-134a: (a) Before identification and (b) after identification.

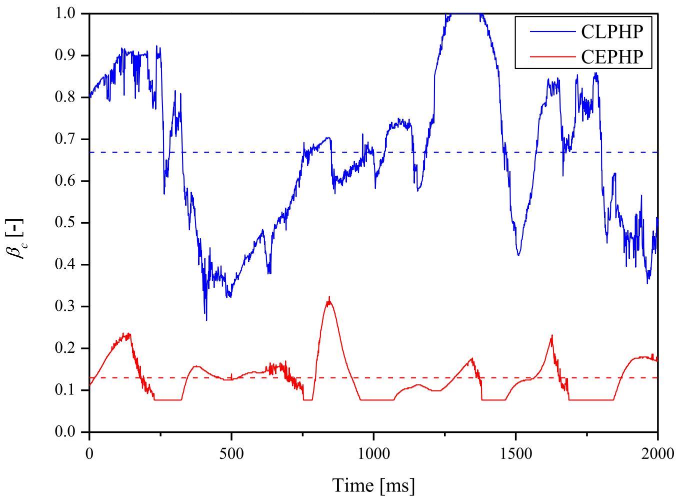

Fig. 11. Volumetric fraction in the condenser section of the CLPHP and the CEPHP filled with R-134a at 24 W.

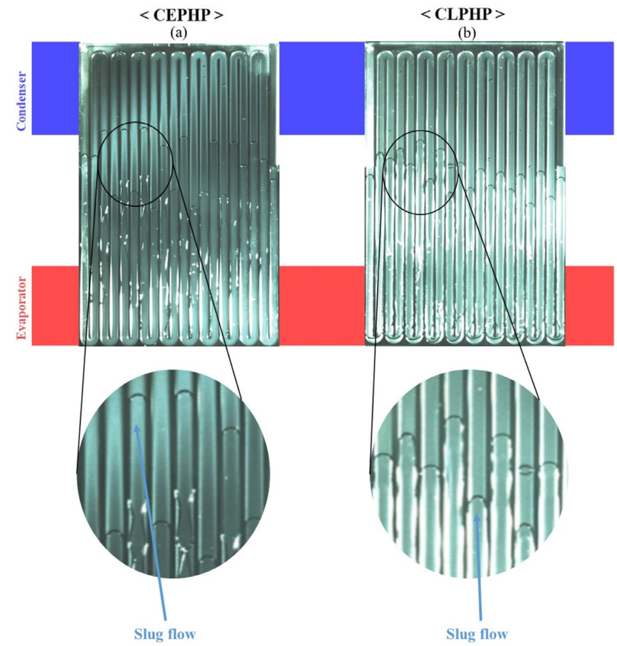

Fig. 12. Flow characteristics of (a) the CEPHP and (b) the CLPHP filled with ethanol at 24 W.

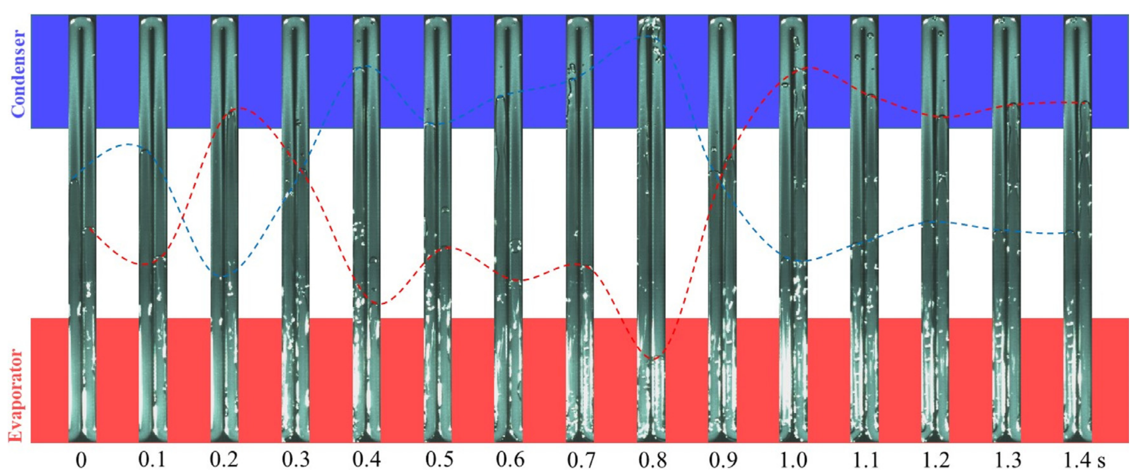

Fig. 13. Temporal variations of liquid slugs and vapor plugs in the 10th and 11th channels of the CEPHP filled with ethanol at 24 W.

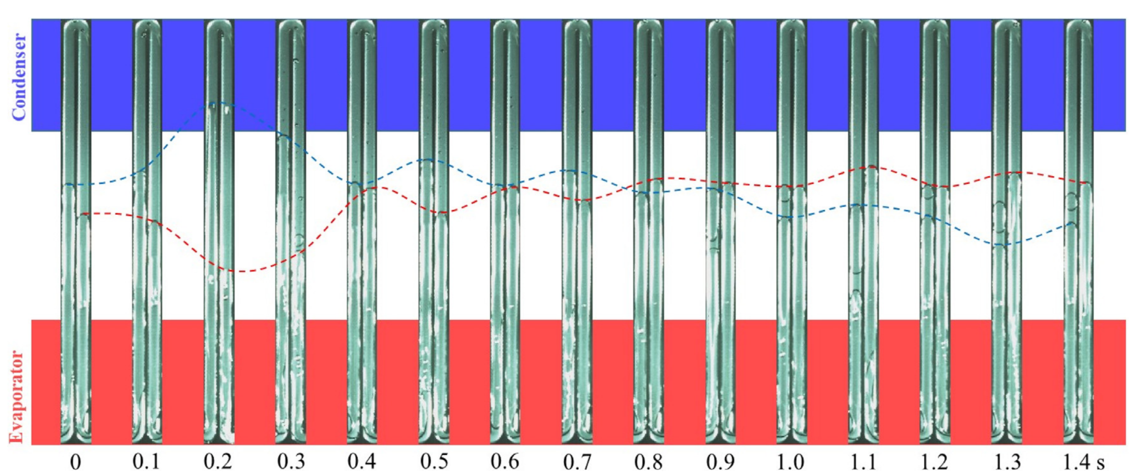

Fig. 14. Temporal variations of liquid slugs and vapor plugs in the 10th and 11th channels of the CLPHP filled with ethanol at 24 W.

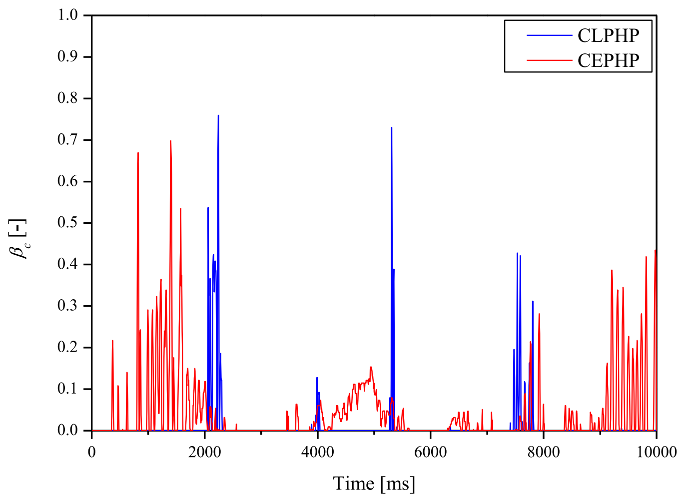

Fig. 15. Volumetric fraction in the condenser section of the CLPHP and the CEPHP filled with ethanol at 24 W.

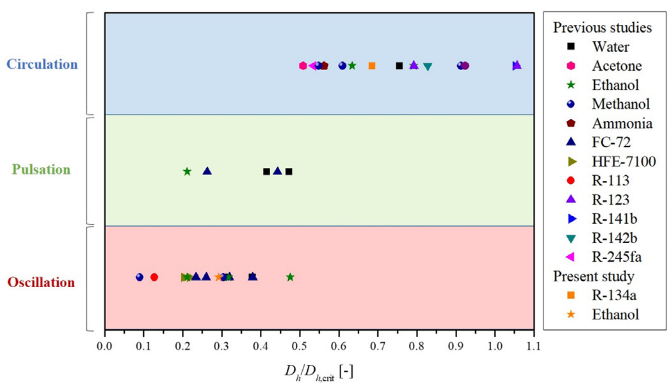

Fig. 16. Classification of a flow behavior inside a CLPHP depending on the $D_h/D_{h,\text{ crit}}$ with various working fluids.

volumetric fraction in the condenser section, β_c is defined as follows:

$$\beta_c \equiv \frac{V_{vp,c}}{V_c} = \frac{A_{vp,c}}{A_c} \tag{3}$$

where $V_{vp,\ c}$, V_c, $A_{vp,\ c}$, and A_c are the volume occupied by vapor plugs in the condenser section, the total volume of the channel in the condenser section, the area occupied by vapor plugs in the condenser section, and the total area of the channel in the condenser section, respectively. Fig. 10 illustrates the identification of the areas occupied by vapor plugs and liquid slugs in the 10th and 11th channels of the CLPHP. Before identifying the vapor plugs and liquid slugs, the original snapshots of the 10th and 11th channels of the CLPHP are shown in Fig. 10(a). The red lines in Fig. 10(a) represent the vapor-liquid menisci. Fig. 10(b) depicts the snapshots of the 10th and 11th channels of the CLPHP after the identification process is completed. The pink-colored area and the green-colored area in Fig. 10(b) represent the area occupied by liquid slugs and vapor plugs, respectively. The identification process is similar to that used in Jo et al. [16]. From the above identification, the volumetric fraction in the 10th and 11th channels of the condenser section of both types of PHPs at the input power of 24 W are evaluated, as shown in Fig. 11. At the input power of 24 W, the average volumetric fraction in the condenser section of the CEPHP and that of the CLPHP are 0.13 and 0.67, respectively. In the case of the CLPHP, the average volumetric fraction in the condenser section of the vapor plugs is 5.2 times higher than that of the CEPHP. The average volumetric fraction in the condenser section is closely related to heat transfer in the condenser section. In the condenser section, heat transfer can be classified into three types: heat transfer through (i) liquid slugs, (ii) vapor plugs enclosed with saturated liquid films and (iii) vapor plugs enclosed with subcooled liquid films. In this study, the liquid films are assumed to be saturated owing to the same reason reported in earlier work [15]. The time for heating the liquid film to the saturation temperature is much smaller than film-covering time, which indicates that assumption is valid. Therefore, heat transfer through vapor plugs enclosed with subcooled liquid films is not taken into account in the present study. From these reasons, heat transfer through liquid slugs and the liquid films enclosing vapor plugs are regarded as sensible and latent heat transfer, respectively. From this classification, it can be inferred that as the average volumetric fraction in the condenser section increases, the overall contribution of latent heat transfer in the condenser section is increased. Previous studies reported that the dominant heat transfer mechanism in a PHP is latent heat transfer [16–18]. Moreover, it is generally known that heat transfer through film condensation has a much higher heat transfer coefficient than convective heat transfer through the liquid slugs. Therefore, an increase of the average volumetric fraction in the condenser section enhances the thermal performance of a PHP due to an improvement of latent heat transfer. From these reasons, a CLPHP performs better than a CEPHP when a CLPHP is observed to operate in a circulation mode.

3.2.2. CLPHP operating in an oscillation mode

Fig. 12 shows the flow characteristics of both types of PHPs. In both types of PHPs, churn/annular flow is not observed as the flow pattern of vapor plugs. Only slug flow is observed as the flow pattern of vapor plugs inside both types of PHPs as depicted in Fig. 12. Furthermore, Figs. 13 and 14 present the temporal variations of liquid slugs and vapor plugs in the 10th and 11th channels of the CEPHP and that of the CLPHP, respectively. The blue and red dashed lines in Figs. 13 and 14 represent the positions of the menisci in the 10th and 11th channels of both types of PHPs. Both types of PHPs are observed to operate in an oscillation mode, as reported by Yoon and Kim [12]. As depicted in Figs. 13 and 14, the CEPHP shows a larger amplitude of oscillating flow compared to the CLPHP. Therefore, vapor plugs inside the CEPHP penetrate deeply into the condenser section compared to the vapor plugs inside the CLPHP. To quantify the penetration rate of vapor plugs,

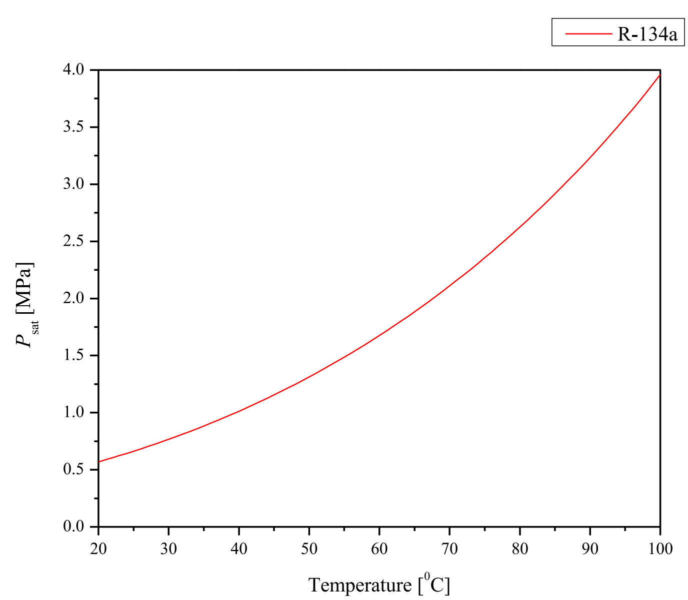

Fig. A1. The variation of P_{sat} with temperature for R-134a.

Table B1
Detailed parameters of flow visualization data.

Researcher	Working fluid	Hydraulic diameter [mm]	$D_h/D_{h,\text{ crit}}$ [-]	Number of turns	Substrate material	Flow behavior
Yoon et al. ([19]	Water	1.65	0.31	6	Copper	Oscillation
Nagwase and Pachgare [20]	Water	2	0.38	2	Glass	Oscillation
Saha et al. [21]	Water	4	0.76	1	Glass	Circulation
Xu et al. [22]	Water	2	0.38	4	Glass	Oscillation
Leu and Wu [23]	Water	2.2 (Surface modified)	0.42	3	Glass	Pulsation
Wilson et al. [24]	Water	1.65	0.31	6	Copper	Oscillation
Lin et al. [25]	Water	1.3	0.25	4	Silicone	Oscillation
Karthikeyan et al. [26]	Water (Nanofluid)	2.0	0.47	5	Copper	Pulsation
Wilson et al. [24]	Acetone	1.65	0.51	6	Copper	Circulation
Mameli et al. [27]	Ethanol	2	0.63	2	Glass	Circulation
Kim et al. [28]	Ethanol	1.5	0.48	4	Brass	Oscillation
Sun et al. [29]	Ethanol	2.5	0.79	7	Glass	Circulation
Yang et al. [30]	Ethanol	1	0.32	33	Aluminum	Oscillation
Yang et al. [30]	Ethanol	2	0.63	20	Aluminum	Circulation
Kwon and Kim [31]	Ethanol	0.75/0.5 (Dual diameter)	0.21	5	Silicon	Pulsation
Jun and Kim [11]	Ethanol	0.667	0.21	5/10/15/20	Silicon	Oscillation
Yoon and Kim [32]	Ethanol	0.667	0.21	5	Silicon	Oscillation
Kim and Kim [33]	Ethanol	0.667	0.21	10	Silicon	Oscillation
Kim and Kim [15]	Ethanol	0.667	0.21	5	Silicon	Oscillation
Xu et al. [22]	Methanol	2	0.61	4	Glass	Circulation
Tong et al. [34]	Methanol	1.8	0.55	7	Glass	Circulation
Liu et al. [13]	Methanol	1	0.3	10	Glass	Oscillation
Liu et al. [13]	Methanol	2/3	0.61/0.91	10	Glass	Circulation
Yang et al. [35]	Methanol	0.33/0.27 (Dual diameter)	0.09	32	Silicon	Oscillation
Xue and Qu [36]	Ammonia	2	0.56	6	Glass	Circulation
Qu et al. [37]	FC-72	0.352	0.23	5	Silicon	Oscillation
Qu et al. [37]	FC-72	0.394	0.26	5	Silicon	Pulsation
Lee and Kim [38]	FC-72	0.39/0.48/0.57	0.26/0.32/0.38	5	Silicon	Oscillation
Kwon and Kim [31]	FC-72	0.75/0.5 (Dual diameter)	0.44	5	Silicon	Pulsation
Sun et al. [39]	HFE-7100	0.357	0.20	11	Silicon	Oscillation
Yang et al. [40]	HFE-7100	0.381	0.22	4	Silicon	Oscillation
Qu et al. [37]	R-113	0.251	0.13	5	Silicon	Oscillation
Soponpongpipat et al. [41]	R-123	1.5/2	0.79/1.06	2	Glass	Circulation
Kim et al. [28]	R-142b	1.5	0.83	10	Brass	Circulation
Spinato et al. [42]	R-245fa	1	0.54	1	Glass	Circulation
Present study	R-134a	0.923	0.69	10	Silicon	Circulation
Present study	Ethanol	0.923	0.29	10	Silicon	Oscillation

the volumetric fraction in the 10th and 11th of the condenser section of both types of PHPs is also compared, as shown in Fig. 15. The average volumetric fraction in the condenser section of the CEPHP and that of the CLPHP are 0.04 and 0.01, respectively. Unlike the results of PHPs filled with R-134a, the average volumetric fraction in the condenser section of the CEPHP is higher than that of the CLPHP. Therefore, owing to the same reason mentioned in Section 3.2.1, a CEPHP performs better than a CLPHP when an oscillation mode is observed inside a CLPHP.

3.3. Criterion for determining the better PHP between a CLPHP and a CEPHP

To figure out a better type of PHP between a CLPHP and a CEPHP, a criterion should be suggested to predict whether a circulation mode or an oscillation mode will be observed inside a PHP. It is postulated that a flow behavior inside a CLPHP in a vertical orientation under bottom heating condition will change from an oscillation mode to a circulation mode as $D_h/D_{h,\text{ crit}}$ increases, which is dealt with in Section 2.1. Based on the postulate, the relation between $D_h/D_{h,\text{ crit}}$ and the flow behavior is further investigated. Using the flow visualization data of the CLPHP in the literature [11,13,15,19–42], the flow behavior inside a CLPHP is classified in terms of a function of $D_h/D_{h,\text{ crit}}$ for various working fluids, as shown in Fig. 16. Some of the flow visualization results cannot be classified as either a circulation mode or an oscillation mode, because those results are presented as an oscillating flow superposed with a circulating flow. Therefore, such kind of flow visualization results is classified as a pulsation mode. The detailed parameters of the flow visualization data are summarized in Appendix B. Regardless of the working fluid, a flow behavior inside a CLPHP changes from an oscillation mode to a circulation mode when $D_h/D_{h,\text{ crit}}$ exceeds 0.5. The experimental results of the present study match well with the above results. The CLPHP filled with ethanol ($D_h/D_{h,\text{ crit}}$= 0.29) is observed to operate in an oscillation mode, while the CLPHP filled with R-134a ($D_h/D_{h,\text{ crit}}$= 0.69) is observed to operate in a circulation mode. Based on the findings, a simple guideline can be suggested to determine a better PHP between a CLPHP and a CEPHP. A CEPHP performs better than a CLPHP when $D_h/D_{h,\text{ crit}}$ is lower than 0.5, while a CLPHP performs better than a CEPHP when $D_h/D_{h,\text{ crit}}$ is higher than 0.5. The experimental results reported in the previous studies also show good agreement with the suggested criterion, in addition to the results of the present study. In the case of Jun and Kim [11] ($D_h/D_{h,\text{ crit}}$= 0.21) and the present study ($D_h/D_{h,\text{ crit}}$= 0.29), the CEPHP performs better than the CLPHP. On the contrary, in the case of Yang et al. [10] ($D_h/D_{h,\text{ crit}}$= 0.53) and the present study ($D_h/D_{h,\text{ crit}}$= 0.69), the CLPHP performs better than the CEPHP.

4. Conclusion

In the present study, the effect of a flow behavior inside a CLPHP is investigated to determine which type of a PHP performs better between a CLPHP and a CEPHP. Using MEMS techniques, ten-turn CLPHPs and CEPHPs are fabricated onto a silicon wafer. To allow for flow visualization, Pyrex glass is bonded on top of the etched silicon wafer. The PHPs are filled with either R-134a or ethanol. Based on the experimental data, a criterion of a diameter ratio ($D_h/D_{h,\text{ crit}}$) is suggested to consider the effect of a flow behavior inside a CLPHP when two types of PHPs are compared. The PHP having the higher average volumetric fraction in the condenser section performs better than the other type of PHP regardless of which type of PHP performs better. As the average volumetric fraction in the condenser section increases, latent heat transfer in the condenser section is increased. This results in an increase of the maximum allowable heat flux as well as a decrease of the thermal resistance.

(a) A CEPHP performs better than a CLPHP when $D_h/D_{h,\text{ crit}}<$ 0.5. When the working fluid is ethanol ($D_h/D_{h,\text{ crit}}=$ 0.29), the CEPHP has 33% lower thermal resistance and 20% higher maximum allowable heat flux than the CLPHP. In this case, both types of PHPs are observed to operate in an oscillation mode. The CEPHP has a higher average volumetric fraction in the condenser section than the CLPHP because the CEPHP shows a larger amplitude of oscillating flow compared to the CLPHP.

(b) A CLPHP performs better than a CEPHP when $D_h/D_{h,\text{ crit}}>$ 0.5. When the working fluid is R-134a ($D_h/D_{h,\text{ crit}}=$ 0.69), the CLPHP has 72% lower thermal resistance and 117% higher maximum allowable heat flux than the CEPHP. In this case, the CLPHP is observed to operate in a circulation mode, while the CEPHP is observed to operate in an oscillation mode. The CLPHP has a higher average volumetric fraction in the condenser section than the CEPHP when a circulation mode is observed inside the CLPHP.

Declaration of Competing Interest

None.

Appendix A. Variation of sat *P* with temperature for R-134a

Fig. A1

Appendix B. Classification of the flow behavior inside a CLPHP

Table B1

CRediT authorship contribution statement

Wookyoung Kim: Conceptualization, Validation, Formal analysis, Investigation, Resources, Writing - original draft, Writing - review & editing. **Sung Jin Kim:** Writing - review & editing, Supervision, Project administration, Funding acquisition.

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International Journal of Heat and Mass Transfer 151 (2020) 119439

Contents lists available at ScienceDirect

International Journal of Heat and Mass Transfer

journal homepage: www.elsevier.com/locate/hmt

Corrigenda

Corrigendum to “Effect of a flow behavior on the thermal performance of closed-loop and closed-end pulsating heat pipes” [International Journal of Heat and Mass Transfer 149 (2020) 119251]

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The authors regret that the acknowledgement section was unintentionally omitted in the manuscript. The acknowledgement section is included below:

Acknowledgment

This work was supported by the National Research Foundation of Korea (NRF) grant funded by the Korea government (MSIT) (No. 2012R1A3A2026427). Additional funding was supplied by the U.S. Office of Naval Research (ONR) and ONR Global (N62909-18-1-2028). Dr. Mark Spector and Dr. Sung-Eun Kim were the grant monitors.

The authors would like to apologise for any inconvenience caused.

DOI of original article: 10.1016/j.ijheatmasstransfer.2019.119251

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https://doi.org/10.1016/j.ijheatmasstransfer.2020.119439